

Karishma Kamble

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EDUCATION

University of Illinois Chicago

Master of Science in Business Analytics – GPA 3.8

Chicago, IL

December 2024

Institute of Chemical Technology, Mumbai

Bachelor of Technology in Chemical Technology

Mumbai, India

May 2018

PROFESSIONAL EXPERIENCE

Deloitte

Data Analyst

Chicago, Illinois

February 2025 - Present

- Consolidated and optimized customer and claims data pipelines by integrating SQL, Python, and advanced analytics, combining data from 15+ sources and reducing manual analysis time by 70%, empowering cross-functional teams in both the retail and insurance industries with real-time Power BI dashboards.
- Architected machine learning and NLP solutions including fraud prediction, customer segmentation, and product recommendation engines, boosting online sales by 18%, increasing average order value by 12%, and enabling investigators to prioritize over 300 high-risk insurance claims for early fraud intervention.
- Developed interactive executive dashboards to visualize key KPIs-such as conversion rates, revenue impact, fraud trends, and churn risk-enhancing data-driven decision-making for marketing, risk, and leadership teams across global business units in both retail and insurance sectors.
- Leveraged unsupervised learning and sentiment analysis to extract actionable insights from customer reviews and claim descriptions, driving a 25% increase in campaign response rates and improved detection of subtle fraudulent behaviors in the insurance industry.
- Collaborated closely with marketing, merchandising, and IT stakeholders throughout project lifecycles, ensuring seamless integration, stakeholder alignment, and widespread adoption of analytics solutions across diverse business functions, including insurance.

Kline & Company

Business Data Analyst

Hyderabad, India

August 2022 - July 2023

- Orchestrated comprehensive business process optimization strategy by implementing Python and SQL automation frameworks across three major enterprise sector clients that improved operational efficiency by 15% and reduced costs by 10%, streamlining workflows for over 40 cross-regional stakeholders through advanced data analytics methodologies.
- Architected predictive forecasting models using Python and Excel advanced algorithms that achieved 80% forecast accuracy, enabling strategic supply chain and inventory planning for production balancing, capacity planning, and quarterly sales targets across three major enterprise sector clients, reducing forecast errors by 25% compared to previous manual methods through statistical modeling and regression techniques.
- Engineered automated Tableau dashboard ecosystem with integrated SQL pipelines to digitize manual reporting processes and visualize demand analysis, driving 15% increase in repeat business and 20% improvement in quarterly contract renewals, providing real-time market insights for clients through business intelligence architecture.
- Implemented comprehensive A/B testing framework by conducting 1,000+ stakeholder interviews across supply chain, sales, and user segments and testing survey formats and campaign strategies, boosting marketing effectiveness by 25% and survey response rates by 20%, establishing the company's first formal testing protocols adopted company-wide within 6 months through statistical analysis and stakeholder collaboration.
- Collaborated with senior stakeholders across marketing and operations departments by delivering fact-based demand forecasting and analysis presentations that influenced strategic initiatives and earned positive leadership feedback, strengthening cross-departmental decision-making processes through executive reporting frameworks and comparative data storytelling.

Wanstats Research and Media Pvt. Ltd.

Data Market Research Analyst

Pune, India

March 2020 - July 2022

- Orchestrated end-to-end analytics project management by successfully delivering 50+ research projects over 2 years aligned with business objectives and client requirements, demonstrating consistent project execution capabilities and maintaining high deliverable quality through strategic project coordination and systematic workflow management.
- Developed comprehensive research methodology framework using primary and secondary data sources across 8 industry verticals, translating complex findings into actionable strategic insights that improved client business strategies and reduced client decision-making time, through advanced data interpretation techniques.
- Implemented rigorous QA protocols and process mapping systems to maintain 95% data quality across all research projects and achieve high client retention rates, resolving remaining 5% quality issues through comprehensive post-sales support and client feedback integration, establishing systematic quality assurance methodologies and responsive issue resolution processes.

- Supervised high-performance analytics team of 3 analysts while serving as primary client liaison for 12 major accounts, achieving 60% contract renewal rate and 25% project scope expansion on \$5,000+ baseline projects through client trust and satisfaction, developing team capabilities that improved productivity by 40% through faster turnaround times and reduced rework errors via strategic leadership and mentoring.
- Architected comprehensive dashboard and pipeline ecosystem by building 15 interactive Tableau dashboards and automating 20 data processing workflows in Excel, reducing processing time by 30% and improving accuracy by 15%, enabling real-time reporting capabilities that supported \$5M in client decision-making tracked through dashboard usage monitoring and public market developments validation via automation and visualization technologies.

Future Market Insights

Business Research Analyst

Pune, India

July 2018 - January 2020

- Delivered comprehensive ad-hoc reporting solutions by creating dynamic Excel dashboards for executive stakeholders as part of an elite 20-member fast-track team, empowering faster data-driven decision-making for high-priority time-sensitive client projects where accuracy and speed were critical, demonstrating exceptional performance and reliability through advanced Excel business intelligence development.
- Engineered advanced forecasting models in Excel using statistical algorithms and regression analysis, reducing forecast turnaround time by 50% and maintaining reliable internal database accuracy through MAPE methodology validation, supporting leadership in identifying demand across product lines and geographies to reduce expansion risk and prioritize optimal market growth opportunities through predictive analytics and data modeling.
- Applied sophisticated analytics methodologies including regression analysis, customer segmentation, trend and competitive analysis to client datasets, increasing client conversion rates by 25% and achieving strong client satisfaction demonstrated through repeat business with individual clients engaging up to 4 times annually through advanced statistical techniques and data-driven insights.
- Architected scalable data processing infrastructure to handle 3M+ data points across 60 market segments, improving production speed by 40% and reducing processing errors including mismatched records and missing data for more reliable outputs, enabling analysis of previously inaccessible large and complex datasets that unlocked new customer segments and market opportunities through big data architecture and optimization techniques.
- Recognized with \$3,000 performance excellence award for outstanding contributions to Fast Track Report Team, delivering critical market intelligence that supported client strategic decisions and doubled team report capacity from 3-4 monthly reports to 8 monthly reports while reducing turnaround time from 1 month to 1 week through exceptional analytical performance and streamlined processes.

PROJECTS

Airline On-Time Performance Analytics & Competitor Benchmarking (Alaska Airlines)

December 2024

- Engineered data pipeline and interactive Tableau dashboards analyzing 6M+ BTS flight records to benchmark Alaska Airlines' on-time performance against competitors, identifying high-delay hubs and operational performance gaps, enabling data-driven efficiency recommendations through statistical analysis and visualization.

LLM-based Agent Development Capstone

December 2024

- Architected custom multi-agent LLM automation system using LangGraph, OpenAI, SQL, and Streamlit to streamline vehicle valuation data retrieval from multiple websites, achieving 95% validated accuracy and reducing processing time from 10 minutes to under 1 minute, creating a successfully piloted prototype now being scaled into the company's enterprise-wide valuation platform through advanced AI development and automation architecture.

Sentimental Analysis - Reddit Quit-Smoking Community

August 2024

- Built behavioral analytics pipeline using Hugging Face LLMs, VADER sentiment scoring, and thematic analysis to process 22,000 quit-smoking community posts, improving quit-success prediction accuracy by 40% and delivering insights that informed targeted intervention strategies for 27,000+ community members through machine learning and natural language processing.

PUBLICATIONS

Kamble, K. AI-Driven LLM Agent System

March 2025

AIRRES, 2025, Book of Abstracts, American-CSE.

TECHNICAL SKILLS

Programming & Analytics: SQL, Python, R, Statistical Modeling, Machine Learning, Predictive Analytics, NLP, Text Mining, Data Mining.

Business Intelligence & Visualization: Tableau, Power BI, Looker, Excel, VBA, Pivot Tables, Power Query, DAX, Data Visualization, Dashboard Development.

Cloud & Data Engineering: AWS, Microsoft Azure, ETL Pipelines, Data Warehousing, Big Data Analytics, Spark, Hadoop.

AI & Machine Learning: TensorFlow, XGBoost, NLP libraries, Random Forest, Logistic Regression, Generative AI, LangChain, LangGraph, AutoGen, Predictive Modeling, Forecasting, Scikit-learn, Hugging Face, OpenAI.

Database & Tools: Database Management, Alteryx, JIRA, Git, Salesforce CRM, Statistical Software

Core Competencies: Stakeholder Communication, Requirements Gathering, Process Improvement, Data Storytelling, Cross-Functional Collaboration, Agile/Scrum, Business Analysis, Project Management.